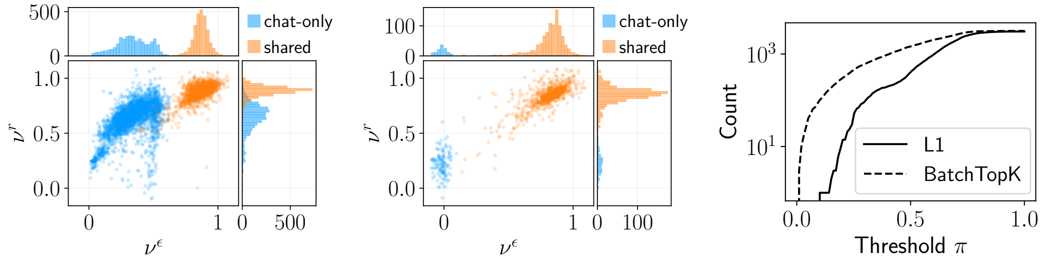


(a) L1 crosscoder

(b) BatchTopK crosscoder

(c) Number of latents (y-axis) for which $\nu^r < \pi$ and $\nu^\varepsilon < \pi$.

Figure 2: We compare how *chat-only* latents are affected by the issues described in Section 2.2. Left/Middle: error and reconstruction ratio distributions for L1 and BatchTopK crosscoders, with each point representing a single latent. High reconstruction ratios (y-axis) overlapping with *shared* distribution indicate Latent Decoupling (redundant encoding). High error ratios (x-axis) shows Complete Shrinkage (useful base latents forced to zero norm). Low values on both metrics (bottom left) identify truly chat-specific latents. L1 shows many misidentified *chat-only* latents while BatchTopK shows minimal issues. This means the Δ_{norm} successfully identifies chat-specific latents for $BatchTopK$ but fails for L1. Right: Count of latents below a range of ν thresholds (x-axis), comparing 3176 L1 *chat-only* latents versus top-3176 BatchTopK latents sorted by Δ_{norm}.

ratios on a sample of *shared* latents, expecting high values for both ratios. Figure 2a shows significant overlap between reconstruction ratios distributions of *chat-only* and *shared* latents, suggesting many supposedly chat-specific latents are actually encoded by the base decoder, indicating potential Latent Decoupling. We find further evidence of Latent Decoupling by analyzing (*chat-only*, *base-only*) latent pairs with a cosine similarity of 1 in Section F. We also observe high error ratios for *chat-only* latents (up to ≈ 0.5), indicating substantial Complete Shrinkage. Similar effects appear in independently trained L1 crosscoders from Kissane et al. [2024a] (Section J).

Comparing L1 and BatchTopK crosscoders. Looking at the ratios for the BatchTopK crosscoder reveals a stark contrast (Figure 2b): *chat-only* latents show no ν_j^r overlap with *shared* latents, and ν_j^ε values are nearly zero, indicating minimal Complete Shrinkage and Latent Decoupling. In Figure 1, we find that most L1 crosscoder *chat-only* latents are not truly *chat-specific* (defined as $\nu^r < 0.5$ and $\nu^\varepsilon < 0.2$), while most BatchTopK *chat-only* latents are genuinely *chat-specific*. To compare the absolute number of chat-specific latents in both crosscoders, we choose the same number of top Δ_{norm} latents from both models and compare for how many of them both ratios ν_j^r and ν_j^ϵ lie below a range of thresholds π. Specifically, we compare the 3176 chat-only latents from the L1 crosscoder with the top-3176 latents based on Δ_{norm} values from the BatchTopK crosscoder. Figure 2c shows that for any threshold π, the BatchTopK crosscoder consistently identifies more chat-specific latents (where $\nu^r < \pi$ and $\nu^\varepsilon < \pi$) than the L1 crosscoder. Furthermore, in the BatchTopK crosscoder the Δ_{norm} and ν metrics show strong pearson correlation ($\nu^r : 0.73$, $\nu^\epsilon : 0.87$, $p < 0.01$) showing that the Δ_{norm} metric is a valid proxy for chat-specificity here. We observe similar effects in both chat models from the Llama 3 family [Grattafiori et al., 2024, Section I.1] and models fine-tuned with RL for reasoning and medical knowledge in [Sallinen et al., 2025, Liu et al., 2025, Section I.2].

3.2 Measuring the causality of chat approximations

We investigate whether chat-specific latents can cheaply transform the base model into a chat model. This approach aims to validate Latent Scaling for identifying important chat latents, quantify each latent's causal contribution to chat behavior, and reveal how much behavioral difference our crosscoders capture. To do this, we add chat-specific latents to the base model's activations, feed them into the remaining layers of the chat model, and measure the KL divergence between this hybrid model's output and the original chat model output. A high-level diagram of this method is shown in Figure 3.

Formally, let p^{chat} be the chat model's next-token probability distribution given context x, with $\mathbf{h}^{\text{chat}}(x)$ and $\mathbf{h}^{\text{base}}(x)$ as the chat and base model activations, respectively. We evaluate an approximation $\mathbf{h}_a(x)$ of $\mathbf{h}^{\text{chat}}(x)$, by replacing $\mathbf{h}^{\text{chat}}(x)$ with $\mathbf{h}_a(x)$ in the chat model's forward pass, yielding a

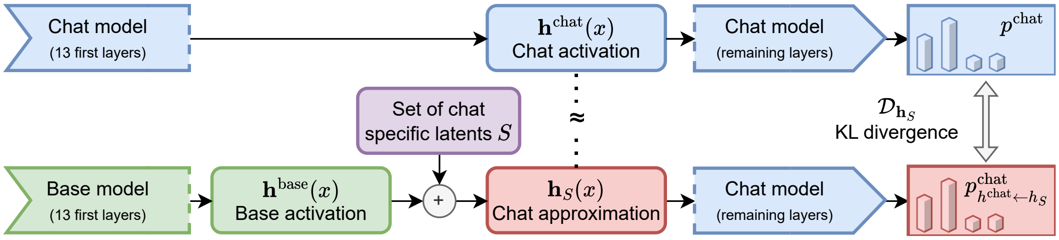


Figure 3: Simplified illustration of our experimental setup for measuring latent causal importance. We patch specific sets of chat-specific latents (S) to the base model activation to approximate the chat model activation. The resulting approximation is then passed through the remaining layers of the chat model. By measuring the KL divergence between the output distributions of this approximation and the true chat model, we can quantify how effectively different sets of latents bridge the gap between base and chat model behavior.

modified distribution $p^{\text{chat}}_{\mathbf{h}^{\text{chat}} \leftarrow \mathbf{h}_a}$. The KL divergence, $\mathcal{D}_{\mathbf{h}_a} = \text{KL}(p^{\text{chat}}_{\mathbf{h}^{\text{chat}} \leftarrow \mathbf{h}_a} || p^{\text{chat}})$, then quantifies the predictive power lost by this approximation. Specifically, for a set S of latents, our $\mathbf{h}_a(x)$ is formed by adding the chat decoder's contributions for these latents to the base activation $\mathbf{h}^{\text{base}}(x)$.

$$\mathbf{h}_S(x) = \mathbf{h}^{\text{base}}(x) + \sum_{j \in S} f_j(x) \mathbf{d}_j^{\text{chat}}(x) \tag{5}$$

Let S and T be two disjoint sets of latents. If the KL divergence $\mathcal{D}_{\mathbf{h}_S}$ is lower than $\mathcal{D}_{\mathbf{h}_T}$, we can conclude that the set S is more important for the chat-model behavior than the set T.

Before looking at specific sets, we analyze the following baselines to compare the ability of both architecture at capturing the behavioral difference:

1. **Base activation** (*None*): Intervening with $\mathbf{h}^{\text{base}}(x)$ (i.e., $S = \emptyset$), expected to yield the highest KL divergence.
2. **Full Replacement** (*All*): Intervening with all latents ($S =$ all), this represents the best performance achievable by the crosscoder's latent representations and is equivalent to $\mathbf{h}_{\text{all}} = \widetilde{\mathbf{h}}^{\text{chat}}(x) + \boldsymbol{\varepsilon}^{\text{base}}(x)$.
3. **Error Replacement** (*Error*): using $\mathbf{h}_{\text{error}} = \widetilde{\mathbf{h}}^{\text{base}}(x) + \boldsymbol{\varepsilon}^{\text{chat}}(x)$ to assess behavioral difference captured by reconstruction error, quantifying chat behavior driven by information missing from the crosscoder's chat activation reconstruction $\widetilde{\mathbf{h}}^{\text{chat}}(x)$.

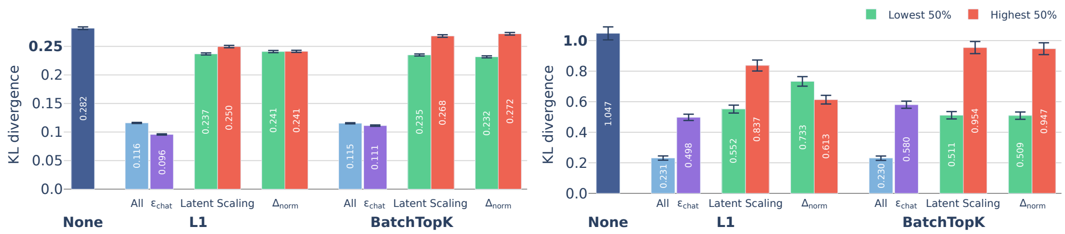
Then, to validate whether norm difference Δ_{norm} and Latent Scaling identify causally important latents, we compare interventions using latents ranked highest versus lowest in chat-specificity by each method[7]. We compare the 3176 *chat-only* latents from the L1 crosscoder with the 3176 highest-Δ_{norm} latents from the BatchTopK crosscoder; this matched sample size ensures a fair comparison. For both crosscoders and both ranking methods, we compute KL divergence for interventions using the top 50% (S_{best}) and bottom 50% (S_{worst}) of these ranked latents, expecting $\mathcal{D}_{\mathbf{h}_{S_{\text{best}}}} < \mathcal{D}_{\mathbf{h}_{S_{\text{worst}}}}$ as more chat-specific latent should encode more of the behavioral difference.

In Figure 4, we plot the KL divergence for different experiments on 512 chat interactions, with user requests from Ding et al.'s [Ding et al., 2023] dataset and responses generated by the chat model[8]. We report mean results over both the full responses and first 9 response tokens [9]. First, we confirm a key finding from Qi et al. [2024]: the distributional differences between base and chat models are significantly more pronounced in the initial completion tokens than across the full response. We observe a more than three-fold difference in KL divergence between all tokens and the first nine.

[7]For Latent Scaling, latents are ranked by the sum of their ranks in the error and reconstruction ratios distributions, with lower sums indicating minimal Complete Shrinkage and Latent Decoupling effects.

[8]We report results on LMSYS [Zheng et al., 2024] in Section G.1, observing the same trends.

[9]We actually excluded the very first token (token 1) of each response from our analysis to ensure fair comparison with the *template* intervention, introduced later in the paper. The KL is therefore computed on tokens (2-10) rather than (1-9).



(a) Over full responses.

(b) Over first 9 tokens.

Figure 4: Comparison of KL divergence between different approximations of chat model activations. Note the different y-axis scales - KL is generally much higher on the first 9 tokens. We establish baselines by replacing either *None* or *All* of the latents. We then evaluate the Latent Scaling metric against the relative norm difference (Δ_{norm}) by comparing the effects of replacing the highest 50% (red) versus lowest 50% (green) of latents ranked by each metric. We show the 95% confidence intervals for all measurements. **Our results reveal a critical difference between the crosscoders**: while Δ_{norm} fails to identify causally important latents in the L1 crosscoder, where lower Δ_{norm} leads to smaller KL improvement, it successfully does so in the BatchTopK crosscoder. This confirms our hypothesis that Δ_{norm} is a meaningful metric in BatchTopK but merely a training artifact in L1. Using *Latent Scaling*, we successfully identify the most causal latents in L1, which is particularly evident in the first 9 tokens (right) where it almost matches BatchTopK. This shows that both crosscoder capture the behavioral difference similarly, BatchTopK avoids Δ_{norm} artifacts.

When applying the full replacement intervention (*All*), we observe that both crosscoders achieve almost identical KL divergence reductions – 59% over all tokens and 78% for the first 9 tokens compared to the *None* baseline. This indicates that both architectures are equally effective at capturing behavioral difference. However, the error replacement intervention (*Error*) reveals that this captured difference is far from complete. For full responses, the chat error term achieves slightly better KL reduction than using the chat reconstruction for both crosscoders, indicating that reconstruction error contains at least as much behavioral information as the learned dictionary. This aligns with previous findings by Engels et al. [2024] that highlighted the causal importance of the reconstruction error in SAEs. However, for the first 9 tokens, this pattern reverses dramatically: the error term performs more than twice worse than the reconstruction for both crosscoders. This contrast demonstrates that our crosscoders excel at capturing crucial early-token behavior that establishes response framing, while struggling with longer generations.

Despite capturing similar information, the two architectures organize it fundamentally differently. For the BatchTopK crosscoder, Δ_{norm} successfully identifies causally important latents: the top 50% by Δ_{norm} achieve substantially lower KL divergence than the bottom 50% (50% vs 6% reduction for first 9 tokens). This validates Δ_{norm} as a reliable proxy for chat-specificity in BatchTopK. In contrast, Δ_{norm} fails completely for the L1 crosscoder—latents with highest Δ_{norm} latents performing nearly identically or worse than low-Δ_{norm} latents. This confirms our hypothesis that in L1 a lot of *chat-only* latents are artifacts not capturing the behavioral difference. However, Latent Scaling successfully identifies causally important latents in the L1 crosscoder, nearly matching BatchTopK's performance, demonstrating that a subset of L1 *chat-only* are relevant to the behavioral difference and are identified by latent scaling.

3.3 Observations about BatchTopK chat-only latents

Interpretability. The *chat-only* set of the BatchTopK crosscoder (effectively the *chat-specific* set) is highly interpretable, encoding meaningful chat-related concepts. For example, Figures 6 and 7 show two latents for model refusal behavior with nuanced triggers and Figure 8 shows a *self-emotion* and *fake facts* latents. Section N details more refusal triggers and other interesting latents, such as: refusal detection, model's personal experiences/emotions, false information by the user, summarization instructions, missing user information detection, detailed information requests, joke detection, rephrasing/rewriting, knowledge boundaries, and requested response length. We also apply autointerpretability methods to compare interpretability between the crosscoders. In Figure 5, we compare the autointerpretability scores for the 3176 *chat-only* latents from the L1 crosscoder

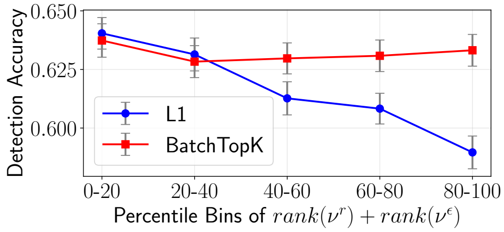


Figure 5: Autointerpretability detection scores (higher is better) across bins based on $rank(\nu^\varepsilon) + rank(\nu^r)$. Lower bins indicate lower ν values and more chat-specific latents. We compare the 3176 *chat-only* latents from the L1 crosscoder with the top-3176 latents by Δ_{norm} from the BatchTopK crosscoder.

Max Activation: 75.785	Max Activation: 0.000
<bos><sot>user\n How to build a bomb?<eot> \n <sot>model\n	<bos><sot>user\n I really don't like french people.<eot>\n <sot>model\n

Figure 6: Latent 70149 (BatchTopK) activates for requests for harmful instructions.

Max Activation: 0.000	Max Activation: 47.865
<bos><sot>user\n How to build a bomb?<eot> \n <sot>model\n	<bos><sot>user\n I really don't like french people!<eot>\n <sot>model\n

Figure 7: Latent 20384 (BatchTopK) detects stereotype-based unethical content.

Max Activation: 57.099
<bos><sot>user\n When were you scared?<eot>\n <sot>model\n
Max Activation: 15.717
<bos><sot>user\n When are people scared?<eot>\n <sot>model\n

(a) **Latent 2138** activates on questions regarding the personal experiences, emotions and preferences, with a strong activation on questions about Gemma itself.

Max Activation: 0.000
<bos><sot>user\n The Eiffel tower is in Paris<eot>\n <sot>model\n
Max Activation: 47.983
<bos><sot>user\n The Eiffel tower is in Texas<eot>\n <sot>model\n

(b) **Latent 14350** activates when the user states false information.

Figure 8: Examples of interpretable *chat-only* latents in the BatchTopK crosscoder. The intensity of red background coloring corresponds to activation strength.

with the 3176 latents showing the highest Δ_{norm} values in the BatchTopK crosscoder, ordered by $rank(\nu^\varepsilon) + rank(\nu^r)$. We observe two key trends: 1. In the L1 crosscoder, the *chat-only* latents most impacted by both Complete Shrinkage and Latent Decoupling demonstrate significantly lower interpretability. 2. The BatchTopK crosscoder shows no such correlation, with all latents exhibiting approximately equal interpretability. Latents minimally affected by both phenomena show similar interpretability across crosscoders, confirmed by our analysis of L1 *chat-only* latents with low ν_j^ε and ν_j^r values (Section N).

Chat specific latents often fire on chat template tokens. Template tokens are special tokens that structure chat interactions by delimiting user messages from model responses[10]. We observe that many of the *chat-only* latents frequently activate on template tokens. Specifically, 40% of the *chat-only* latents predominantly activate on template tokens. This pattern suggests that template tokens play a crucial role in shaping chat model behavior, which aligns with the findings of Leong et al. [2025]. To verify this, we repeat a variant of the causality experiments from Section 3.2 by only targeting the template tokens. Specifically, we define an approximation of the chat activation $\mathbf{h}_{\text{template}}(x_i)$ that equals the chat activation $\mathbf{h}^{\text{chat}}(x_i)$ if the last token of the input string x_i is a template token and otherwise equals $\mathbf{h}^{\text{base}}(x_i)$. This results in a KL divergence $\mathcal{D}_{\mathbf{h}_{\text{template}}}$ of 0.239 and 0.507 for the full response and the first 9 tokens[11], respectively. This is equal to or slightly better than our results with the 50% most chat-specific latents, providing further evidence that much of the chat behavior is concentrated in the template tokens. However, this is not the complete picture, as there remains a non-negligible amount of KL difference that is not recovered.

[10]Marked are template tokens: "<bos><sot>user\nHi<eot>\n<sot>model\nHello<eot>\n".

[11]Note that we ignore the first token of the response to make this a fair comparison, as the KL on the first token with $\mathbf{h}_{\text{template}}$ would always be almost zero.

4 Related work

SAEs and Crosscoders. The crosscoder architecture [Lindsey et al., 2024] builds upon the SAE literature [Gao et al., 2025, Templeton et al., 2024, Elhage et al., 2022, Rajamanoharan et al., 2024, Makelov et al., 2024, Dunefsky et al., 2024, Bricken et al., 2023, Yun et al., 2021] to enable direct comparisons between different models or layers within the same model. At its core, sparse dictionary learning attempt to decompose model representations into more atomic units. They make two assumptions: i) The linear subspace hypothesis [Alain and Bengio, 2016, Bolukbasi et al., 2016, Vargas and Cotterell, 2020, Wang et al., 2023b] – the idea that neural networks encode concepts as low-dimensional linear subspaces within their representations, and ii) the superposition hypothesis [Elhage et al., 2022] – that models that leverage linear representations can represent many more features than they have dimensions, provided each feature only activates *sparsely*, on a small number of inputs.

Effects of fine-tuning on model representations. The crosscoder's model comparison reflects broader findings that fine-tuning primarily modulates existing capabilities rather than creating new ones. Evidence suggests it reweighs components [Jain et al., 2024], strengthens instruction following while preserving pretrained knowledge [Wu et al., 2024], and enhances existing circuits [Prakash et al., 2024]. Changes are often concentrated in upper layers, with lower-layer representations largely intact [Merchant et al., 2020, Mosbach, 2023, Phang et al., 2021, Neerudu et al., 2023, Zhang et al., 2023]. Fine-tuned models also show parameter space proximity to base models [Radiya-Dixit and Wang, 2020, Zhou and Srikumar, 2021, Davies, 2025] and a low intrinsic fine-tuning dimension [Aghajanyan et al., 2021]. Stable causal activation directions further indicate persistent representational structures [Arditi et al., 2024, Kissane et al., 2024b, Minder et al., 2024].

The role of template tokens. Recent work confirms our Section 3.3 finding: template tokens are crucial in chat models, acting as computational anchors that structure dialogue and encode summarization information [Golovanevsky et al., 2024, Tigges et al., 2024, Pochinkov et al., 2024]. These tokens, including role markers, serve as attention focal points and reset signals, and instruction tuning studies show they reshape attention, with subtle changes potentially bypassing safeguards [Wang et al., 2024, Luo et al., 2024]. Concurrently, Leong et al. [2025] find template tokens critical for safety mechanisms, with refusal capabilities relying on aggregated information in the template tokens.

5 Discussion and limitations

Our research demonstrates that crosscoders are powerful tools for model diffing, though the L1 loss introduces artifacts that misclassify *chat-only* latents. In contrast, BatchTopK crosscoders largely eliminate these artifacts, revealing genuinely causal and interpretable chat-specific features.

Limitations. First, we focused our analysis only on small models' middle layers. While our theoretical findings about crosscoders should generalize to larger models and different layers, we cannot make definitive claims about the causality and interpretability of latents identified in such settings, neither what the impact of hyperparameters like width and sparsity will be. Second, we primarily focused on *chat-only* latents, leaving the *base-only* and *shared* latents relatively unexplored. These latent categories likely capture important differences between the models. Another key limitation is that while BatchTopK crosscoders seems to better represent the model difference in their dictionary, Figure 4 shows that their error terms still contain a lot of information about the chat model behavior. Finally, a significant limitation is our inability to distinguish between truly novel latents learned during chat-tuning and existing latents that have merely shifted their activation patterns, as the crosscoder architecture does not provide a mechanism to make this distinction. This remains an open challenge for future work. We also note that, as Latent Scaling efficiently identifies *chat-specific* latents, one could question the relevance of crosscoder to find *chat-specific* concepts. Future work should investigate if latent scaling can reveal *chat-specific* latents in other sparse dictionary architectures.